CryptoTrace AI

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TITLE PAGE

Problem Statement ID –
26146

Problem Statement Title –
AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Theme –
Blockchain & Cybersecurity

PS Category –
Software

Team ID –

Team Name (Registered on portal)
CryptoTrace AI

Prototype

Github Repo

About

CryptoTrace AI

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Idea Title

AI-Powered Bitcoin Transaction Intelligence & Investigative Platform

Proposed Solution (Describe your Idea/Solution/Prototype)

CryptoTrace AI is an offline, AI-powered platform that ingests bulk Bitcoin transaction and network metadata, builds an interactive entity graph, detects suspicious activities using machine learning, and provides explainable, prioritized leads for investigators through an intuitive dashboard.

Detailed explanation of the proposed solution

The system collects and processes transaction & network data (CSV/JSON/XML), enriches it using GeoIP/ASN databases, and correlates IPs, wallets, and transactions. It applies AI/ML models (Isolation Forest, DBSCAN) to detect anomalies and clusters. Risk scores and explainable insights are generated and visualized on an investigator dashboard with graphs, timelines, and reports.

How it addresses the problem

It helps investigators trace the flow of illicit funds, uncover hidden relationships, detect unusual patterns, and prioritize high-risk entities efficiently—reducing manual effort, increasing accuracy, and accelerating actionable intelligence.

Innovation and uniqueness of the solution

Combines network-layer metadata with blockchain data, AI-driven anomaly detection, explainable risk scoring, and interactive graph analysis—delivered as a fully offline, privacy-preserving platform for law enforcement and agencies.

SOLUTION WORKFLOW

1. Data Ingestion — CSV / JSON / XML → 2. Entity Correlation — IP → TX → Wallet → 3. Graph Construction — Build Relationship Graph → 4. AI/ML Detection — Anomaly & Clustering → 5. Risk & Explanation — Score & Explain Alerts → 6. Investigator Dashboard — Visualize & Report

Technical Approach

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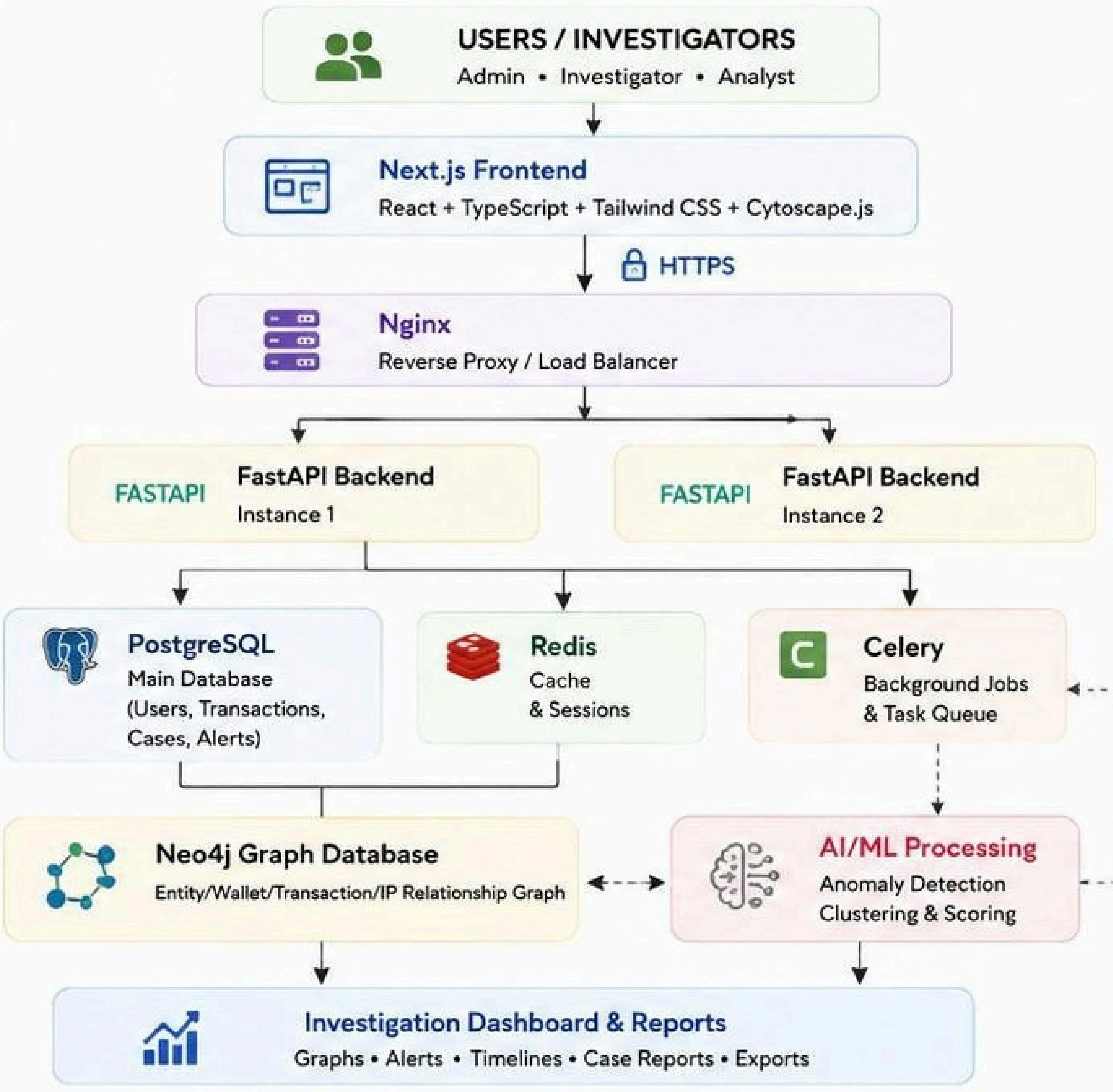
SYSTEM ARCHITECTURE



TECHNOLOGY STACK

Frontend

- Next.js (React Framework)
- TypeScript
- Tailwind CSS
- Cytoscape.js

Interactive UI & Graph Visualization

Backend

- Python
- FastAPI
- Pydantic
- JWT Authentication

High-performance API & Business Logic

Data Layer

- PostgreSQL
- Neo4j Graph DB
- Redis (Cache)
- Pandas / Polars

Structured Data, Graph & High-speed Processing

AI / ML

- Scikit-learn
- Isolation Forest
- DBSCAN
- SHAP

Anomaly Detection, Clustering & Explainability

Processing & Scalability

- Celery (Background Jobs)
- Nginx (Reverse Proxy)
- Docker (Containerization)
- Linux (Offline Deployment)

Scalable, Reliable & Offline Compatible

Security

- HTTPS / TLS
- JWT + RBAC
- Audit Logs
- Encryption

Secure, Private & Auditable System

KEY WORKFLOW

1. Data Ingestion → 2. Correlation → 3. Graph Construction → 4. AI/ML Detection → 5. Risk & Explanation → 6. Dashboard

1. **Data Ingestion** – CSV / JSON / XML Upload & Validation
2. **Correlation** – IP ↔ TX ↔ Wallet Correlation
3. **Graph Construction** – Entity & Transaction Graph Building
4. **AI/ML Detection** – Isolation Forest + DBSCAN
5. **Risk & Explanation** – Risk Scoring + SHAP Explanation
6. **Dashboard** – Visualize • Analyze • Investigate • Report

Goal: Detect suspicious cryptocurrency activities by correlating blockchain and network data using AI/ML and graph analytics, and deliver explainable, prioritized investigative leads.

Offline First | Explainable AI | Scalable & Secure

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Feasibility And Viability

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Feasibility	Challenges / Risks	Mitigation Strategy
Uses open-source technologies such as Python, PostgreSQL, NetworkX and Scikit-learn.	Large datasets may cause slow processing.	Chunked ingestion + background workers + indexing + caching.
Works with the synthetic dataset specified by the problem statement.	IP-to-wallet correlation may be incomplete or noisy.	Treat network correlation as supporting evidence, not definitive identity proof.
Core system can operate offline on Linux.	ML may produce false positives.	Combine ML anomaly scores with graph/ transaction features and human review.
Graph-based analysis is suitable for tracing transaction relationships.	Very large graphs can become difficult to visualize.	Progressive graph expansion and filtering by time/risk/depth.
Modular architecture allows future scaling.	Maintaining explainability of AI decisions.	SHAP + visible evidence/features behind every alert.
Uses containerized services for reproducibility.	Sensitive investigative data requires strong security.	RBAC, encryption, audit logs and isolated offline deployment.

FEASIBLE NOW: Synthetic Dataset → Working Offline Prototype → AI + Graph Detection → Investigator Validation → Future Scale-Up

The system assists investigators by prioritizing suspicious activity and connecting evidence; it does not claim to identify a real-world person solely from a wallet or IP address.

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Impacts And Benefits

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Potential Impact on the Target Audience

Law Enforcement Agencies
Faster identification of suspicious wallets and transactions, stronger evidence collection, and better decision-making.

Financial Institutions & Exchanges
Enhanced AML compliance, reduced risk exposure, and improved monitoring of high-risk transactions.

Investigators & Analysts
Intuitive tools and visualizations reduce manual effort and speed up complex investigation workflows.

Policy Makers & Regulators
Data-driven insights and analytics help in framing better policies and understanding illicit trends.

Citizens & Society
Safer financial ecosystem with reduced cybercrime, fraud, ransomware and money laundering activities.

Benefits of the Solution

Social Benefits
Helps build a safer digital society by tackling financial crimes, protecting victims and increasing trust in financial systems.

Economic Benefits
Reduces financial losses due to fraud, ransomware and money laundering. Helps institutions save compliance and investigation costs.

Operational Benefits
Automates data analysis, reduces investigation time, improves accuracy and enables efficient allocation of investigative resources.

Environmental Benefits
Digital and paperless investigations reduce paper usage and operational overhead.

Legal & Regulatory Benefits
Supports regulatory compliance (AML/CFT), strengthens transparency and aids in legal proceedings.

Technological Benefits
Promotes the adoption of AI, Graph Analytics and Data Science for solving real-world challenges in law enforcement.

CryptoTrace AI empowers investigators with intelligent insights, accelerates detection of illicit activities, and contributes to a secure, transparent and responsible digital economy.

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Research And References

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KEY RESEARCH PAPERS

01

Bitcoin: A Peer-to-Peer Electronic Cash System (2008)

Satoshi Nakamoto

The foundational whitepaper that introduced Bitcoin, its transaction model, timestamping, proof-of-work and peer-to-peer network.

bitcoin.org/bitcoin-paper

02

Detecting Anomalous Cryptocurrency Transactions

Pocher et al., 2023

Uses graph representation and ML (GCN & GAT) techniques to detect anomalous / illicit transactions for AML/CFT applications.

Springer – Electronic Markets

03

Bitcoin Address Clustering Using Multiple Heuristics

He et al., 2022

Clusters Bitcoin addresses into entities using heuristics like common-input ownership, change detection and Louvain algorithm.

IET Blockchain – Wiley

04

Deanonymisation of Clients in Bitcoin P2P Network

Biryukov et al., 2014

Shows how IP addresses can be linked to Bitcoin transactions through network-layer observations and timing.

arXiv: 1405.7418

05

Deanonymization in the Bitcoin P2P Network

Fanti & Viswanath, 2017

Analyzes transaction propagation and shows the risk of linking IP addresses with Bitcoin pseudonyms.

NeurIPS 2017

06

Graph Transformer for Illicit Bitcoin Transaction Detection

Lin et al., 2026

Proposes a wavelet–temporal graph transformer approach achieving high accuracy in detecting illicit transactions.

Scientific Reports (2026)

DATASETS USED

01

Elliptic Bitcoin Transaction Dataset

- 203,769 transaction nodes
- 234,355 edges
- Labeled as licit, illicit, unknown
- 166 features per node
- 49 temporal snapshots

Kaggle – Elliptic Data Set

02

BitcoinHeist Ransomware Dataset

- Jan 2009 – Dec 2018 data
- Address-level behavioral features
- Ransomware + Normal addresses
- ~2.9M instances, 10 features

Kaggle – BitcoinHeist Dataset

03

Bitcoin Blockchain Historical Data

- Raw blockchain data (blocks, transactions)
- Linked with Google BigQuery
- Ideal for building ingestion pipelines & graphs

Kaggle – Bitcoin Blockchain

04

Bitcoin Transactions 2024 Dataset

- 10,000 transaction records
- Fields: timestamp, block, hash, fee, input/output counts, etc.
- Great for testing & feature engineering

Kaggle – Bitcoin TXs 2024

05

Augmented Elliptic Dataset

- Addresses class imbalance in original Elliptic dataset
- Improves model performance and fairness
- Better for training ML models

Kaggle – Augmented Elliptic

TOOLS & REFERENCES

Python, Pandas, NumPy, Scikit-learn
Data processing & Machine Learning

NetworkX, PyG, Cytoscape
Graph construction & visualization

PostgreSQL
Data storage & indexing

GeoIP / ASN DB (MaxMind)
IP enrichment with Geo-location & ASN

SHAP

SHAP
Explainable AI & Feature importance

Bitcoin.org, arXiv, IEEE, Springer, NeurIPS
Research papers & Technical references

These research papers and datasets provide the scientific foundation and real-world data required to build a robust, explainable and accurate Bitcoin transaction intelligence platform.